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Your Name

Your University
your.email@university.edu

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Outline

Introduction

Methodology

Results

Conclusion

Motivation

- **Problem:** Volatility clustering in equity markets
- Traditional models assume constant variance
- This leads to mispriced options and poor risk management

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Research Question

“Does regime-switching improve volatility forecasts over standard GARCH models?”

- Compare GARCH(1,1) vs. Markov-Switching GARCH
- Evaluate using MSE and VaR backtesting

GARCH(1,1) Specification

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (1)$$

Parameter	Definition
ω	Long-run variance
α	Reaction to past shocks
β	Persistence of volatility

Constraint: $\alpha + \beta < 1$ for stationarity

Main Result

Finding #1

MS-GARCH reduces MSE by 18% compared to standard GARCH

- Significant improvement in crisis periods (2008, 2020)
- **Regime 1:** Low volatility (70% of time)
- **Regime 2:** High volatility (30% of time)

Model Comparison

Model	AIC	BIC	MSE
GARCH(1,1)	-4.52	-4.48	0.034
MS-GARCH	-4.81	-4.75	0.028

Lower is better for all metrics

Key Takeaways

1. MS-GARCH outperforms GARCH in crisis periods
2. Regime persistence is high (transition probability $> 95\%$)
3. **Practical implication:** Use MS-GARCH for VaR in turbulent markets

Thank You

Questions? Your Name

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QR code to thesis here (optional)